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Particle Filter Problem

A direct filter method for parameter estimation

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ABSTRACT

Parameter estimation is an important research topic in data assimilation. In this paper, a novel parameter estimation method is introduced, where the parameter is considered as the state process in a nonlinear filtering problem and the state model that contains the parameter is used to construct a pseudo-observation. This approach is named the direct filter method since nonlinear filtering algorithms are used to estimate the parameter directly without estimating the state model as part of the solution in the nonlinear filtering problem. Numerical experiments are carried out to examine the effectiveness and accuracy of the direct filter method.

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1. Introduction

Many mathematical simulations involve parameters that describe the model systems. Determining proper model parameters is a key challenge in mathematical modeling. The major effort in the parameter estimation is to collect observational data of the model system, and then choose appropriate methods to estimate the parameters in the model. With the advances of data collection techniques, parameter estimation has attracted more and more attention and is widely used in many areas, such like physics, chemistry, biology, material science, finance and many engineering disciplines [1–4].

In many real world applications, the mathematical models are dynamical systems and usually perturbed by some noises (known as state-space models). If the parameters are time-varying random variables and can be included in the state space, then any state estimate methods can be applied directly. However, if the true parameters are not varying with time, standard state estimate methods can fail and more sophisticated algorithms are needed. Two well known approaches to estimate the static parameters in noise perturbed dynamical systems are optimization based methods and Bayesian estimation. Popular optimization based methods include maximum likelihood estimation (MLE) [5] and the maximum a posteriori estimate (MAP) [6]. MLE aims to find the parameter values that maximize the likelihood function while MAP minimizes the negative logarithm of the posterior. Different from the point estimate of MLE and MAP, Bayesian estimation treats the parameter as a random variable and fully calculates the posterior distribution of the parameter. In practice, both approaches can be implemented either off-line or on-line. For a long observation sequence, off-line methods are typically prohibitive due to the large amount of data and on-line methods are preferable. In this work, we will concentrate our effort on the Bayesian estimation approach and study on-line parameter estimation as a filtering problem.

Filtering problems identify the state of a system from an uncertain system equation and noisy observations. One class of filtering methods is Kalman filter [7] based methods, which includes Kalman filter, extended Kalman filter, unscented

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Kalman Filter, ensemble Kalman filter, etc. [8–10]. Usually Gaussian assumption is needed in these methods. For general nonlinear non-Gaussian applications, the particle filter, which is also known as "sequential Monte Carlo" [11,12], is one of the most widely accepted methods by practitioners. The particle filter method, and their extended versions, simulate the conditional distribution of the target state by using "particles". However, in order to avoid the degeneracy of the standard particle filters, the number of particles needed will increase exponentially with the dimensions [13]. Therefore, particle filters will fail for high dimensional applications and several advanced particle filters, such as auxiliary particle filter, implicit particle filter, localized particle filter, and many others [14–21], have been developed to improve the performance in high dimensions. Due to the effectiveness and implementability of the filtering algorithms in solving moderate-size real-world problems, and especially in estimating state of systems, the nonlinear filters are extensively studied and applied to parameter estimation for dynamical systems. One common approach using nonlinear filters for parameter estimation is augmenting the state with the parameter [2], in which one transfers the parameter to be estimated to a pseudo parameter process and combine the dynamical system with the parameter process as the augmented process. Then, the combined augmented process is treated as the target state in the filtering problem and the augmented state will be estimated based on observations. As a result of the filtering for the augmented state, the estimations on the parameter would be driven by the estimations on the model state. Therefore, the parameter could be estimated effectively if one has sufficient observational data and accurate enough estimations on the model state. We denote this approach as the augmented filter for parameter estimation and it can be implemented by both Kalman filters and particle filters. In the augmented filter approach for the parameter estimation, if the parameter is considered as a constant process, the parameter space usually cannot be explored adequately and the filter can degenerate [22]. Introducing an artificial dynamic noise to the parameter process turns out to be useful in practice but it may require a significant tuning [2].

In practical applications, estimating the parameter that governs the dynamical system is often the primary problem of interest and the state estimate is not necessary. In this paper, we will develop a novel parameter estimation method by solving a corresponding filtering problem. Instead of combining the dynamical system with the parameter process to form a combined state process as in the augmented filter, in our approach we consider the parameter process as the only target state in the filtering problem. Since the observations are some measurements for the dynamical system, not for the parameters, to implement the filtering algorithms, we introduce a modified observation function to link the parameter with the data. Specifically, we composite the original observation function with the system model function to form a composited observation function that could describe the map from the parameter to the observational data. Due to the fact that we consider the parameter as the only target state in the corresponding filtering problem without estimating the dynamical system, we name our approach the "*direct filter*". To the best of our knowledge, the exploration of compositing the observation with the dynamical system to create a new modified observation function has not yet been discussed before. The novel construction for the observation function leads to one major difference between our direct filter method and the augmented filter – we avoid the challenge of estimating the state of the model dynamics which could be very difficult when the dynamical system is chaotic. This difference would bring us two advantages for the direct filter method compared to the augmented filter: First of all, the direct filter reduces the dimension in the filtering problem significantly by estimating only the parameter process. The dimensionality reduction can improve the efficiency of the filter and mitigate the degeneracy. Second, the direct filter avoids estimating the model state, which allows the nonlinear filter to focus on the accuracy of parameter estimation. However, in order to provide a direct map from the parameter to the observational data, the modified observation function in the direct filter should not contain the state of the dynamical system since it is not available in the algorithm. To achieve this goal, the direct filter needs an "invertible" assumption on the original observation function. We use the observational data that is available to calculate an approximation for the dynamic state by inverting the observation function. This "invertible" assumption is satisfied in many applications which leaves large applicable area for the direct filter method. We note that there's some previous research on pseudo-marginal approach for parameter estimation which allows people to focus more on estimating the parameter (see [15] for an example). However, in those works, the samples for the state variables are still needed in the algorithms, which is the main difference from the direct filter. Here in the direct filter, the samples of the original state variables are avoided totally.

We organize the rest of this paper as follows. In Section 2, we give a brief description for the formulation of parameter estimation problem. Section 3 introduces the nonlinear filter approach for the parameter estimation problem and is followed by our novel direct filter methodology with a specific particle filter implementation. In Section 4, we demonstrate the performance of our algorithm by examining three numerical experiments. We summarize and conclude this paper in Section 5.

2. Problem setting

In this section, we introduce the parameter estimation problem in the state-space model. The state-space model, also known as hidden Markov model, is an important class of time series models. Specifically, a state-space model is composed by two stochastic processes $\{X_n\}_{n\geq 0}$ and $\{Y_n\}_{n\geq 1}$. The process $\{X_n\}_{n\geq 0}$ is a latent Markov process with initial density p_0. In this work, we use the following model to describe the process $\{X_n\}_{n\geq 0}$ as

$$X_{n+1} = h(X_n, \theta) + w_n, \tag{1}$$

where $X_n \in \mathbb{R}^d$ is the state of the dynamical system (1) at time instant n; $h : \mathbb{R}^d \times \mathbb{R}^r \to \mathbb{R}^d$ is a function of X_n and θ, which represents the state model and is determined by a set of parameters in the parameter vector $\theta \in \mathbb{R}^r$; $w_n \in \mathbb{R}^d$ is an additive random noise that perturbs the dynamical system and we assume that for each $n \in \mathbb{N}$, w_n follows a Gaussian distribution with covariance σ, i.e. $w_n \sim N(0, \sigma)$. The goal of the parameter estimation problem is to estimate the parameter vector θ in the state model h and we just call θ the "*parameter*" in the rest of this paper for simplicity. The process $\{X_n\}_{n\geq 0}$ defined in the dynamical system (1) is usually called the "*state*" in the state-space model.

To estimate the parameter θ, we collect observations about the state process. In many applications, the observation we receive is a direct observation $Y_{n+1} \in \mathbb{R}^k$ $(k \geq d)$ about the state X_{n+1}, which is defined as follows

$$Y_{n+1} = HX_{n+1} + \xi_{n+1}, \tag{2}$$

where $H \in \mathbb{R}^{k\times d}$ is an invertible constant matrix applied to the model state X_{n+1}, $\xi_{n+1} \in \mathbb{R}^k$ is another Gaussian noise independent of w_n with covariance γ, i.e. $\xi_{n+1} \sim N(0, \gamma)$, and we call $\{Y_n\}_{n\geq 1}$ the "*observation process*". The state process $\{X_n\}_{n\geq 0}$ together with the observation process $\{Y_n\}_{n\geq 1}$ form the general framework of the state-space model. The goal here is to estimate θ based on the observations $\{Y_n\}_{n\geq 1}$ and computational methods could be generally categorized by two main types: off-line methods and on-line methods [2]. In this paper, we focus on the on-line Bayesian methods [2], which process the observational data as a sequence and update the estimated parameter every time we receive a set of new data Y_n. The on-line methods are quite flexible and would give real-time feedbacks for the parameter estimation.

We define the likelihood function $p(Y_{1:n}|\theta)$ of θ as

$$p(Y_{1:n}|\theta) = \int p(X_{0:n}, Y_{1:n}|\theta) dX_{0:n}, \tag{3}$$

where $p(X_{0:n}, Y_{1:n}|\theta)$ is the joint distribution of the state process $\{X_n\}_{n\geq 0}$ and the observation process $\{Y_n\}_{n\geq 1}$ given a choice of parameter θ, and we use $\lambda_{i:j}$ to denote the sub-sequence $\{\lambda_i, \lambda_{i+1}, \ldots, \lambda_j\}$ for any sequence $\{\lambda_n\}$. In the Bayesian parameter inference, by the Bayes' formula, the conditional probability density of the parameter θ, $p(\theta|Y_{1:n})$ given observational data, is

$$p(\theta|Y_{1:n}) = \frac{p(Y_{1:n}|\theta)p(\theta)}{\int p(Y_{1:n}|\theta)p(\theta)d\theta}, \tag{4}$$

where $p(\theta)$ is the prior probability density of the parameter and $p(Y_{1:n}|\theta)$ is the likelihood defined in (3). The posterior density function $p(\theta|Y_{1:n})$ measures how likely we should choose a parameter given all the observational data $Y_{1:n}$ and contains all the information about the parameter given the observations. However, due to the high dimensional integration involved in (4), direct calculation of useful statistics about θ such as the mean and variance, based on (4), is analytically and computably prohibitive. Monte Carlo methods are widely used and many numerical algorithms are developed to tackle the difficulties. Among those efforts, a popular approach in practice is to implement a recursive data assimilation procedure in which the data is run through once sequentially and the estimated parameter is updated recursively in a sequence. This suggests the application of filtering algorithms, which carry out recursive Bayesian estimations for a dynamical system based on partial observation of the system. In the following section, we discuss the nonlinear filtering algorithm for the parameter estimation problem and introduce our novel direct filter algorithm under the framework of recursive on-line Bayesian parameter inference.

3. Direct filter algorithm for parameter estimation

Before we start the discussion of our direct filter algorithm for the parameter estimation problem, let's first give a brief introduction to the general setup for the nonlinear filtering problem in Subsection 3.1. Then, in Subsection 3.2, we present the augmented concept in filtering algorithms for the parameter estimation problem. This is a common approach when estimating the parameter using filtering algorithms. In Subsection 3.3, we develop our algorithm by introducing a general methodology, and then we provide a specific particle filter implementation of the algorithm in Subsection 3.4.

3.1. Nonlinear filter

We consider the following state-space model of a nonlinear filtering problem,

$$S_{n+1} = F(S_n) + \gamma_n, \tag{5a}$$

$$O_{n+1} = G(S_{n+1}) + \zeta_{n+1}, \tag{5b}$$

where the index $n \geq 0$ represents a time series. Similar to the formulation of parameter estimation problem (1)-(2), $\{S_n\}_{n\geq 0}$ and $\{O_n\}_{n\geq 1}$ in the nonlinear filtering problem (5) are the state process and the observation process with system noise $\{\gamma_n\}_{n\geq 0}$ and observational noise $\{\zeta_n\}_{n\geq 1}$, respectively. The function F describes the dynamical system and G is the observation function. Both F and G could be nonlinear functions. The goal of the nonlinear filtering problem is to obtain the best

estimate of state S_{n+1} given observations $O_{1:n+1}$, which can be expressed as the posterior density of S_{n+1} given $O_{1:n+1}$, i.e. $p(S_{n+1}|O_{1:n+1})$.

The posterior density function $p(S_{n+1}|O_{1:n+1})$ can be estimated in a two-step recursive procedure: a *prediction step* and an *update step*. At time instant n, assuming that we have the conditional probability density $p(S_n|O_{1:n})$, the prediction step uses the knowledge of dynamical system to calculate the prior probability density function (pdf) of the state at time $n+1$ through the Chapman-Kolmogorov formula

$$p(S_{n+1}|O_{1:n}) = \int p(S_n|O_{1:n})p(S_{n+1}|S_n)dS_n, \quad \text{(prediction step)} \tag{6}$$

where $p(S_{n+1}|S_n)$ is the transition density associated with (5a). When the observation information O_{n+1} is available at time instant $n+1$, we apply the Bayes' formula to update the prior pdf and get a posterior pdf $p(S_{n+1}|O_{1:n+1})$ as following

$$p(S_{n+1}|O_{1:n+1}) = \frac{p(O_{n+1}|S_{n+1})p(S_{n+1}|O_{1:n})}{p(O_{n+1}|O_{1:n})}, \quad \text{(update step)} \tag{7}$$

where $p(O_{n+1}|S_{n+1})$ is the likelihood function derived from the observation (5b), and $p(O_{n+1}|O_{1:n}) = \int p(O_{n+1}|S_{n+1})p(S_{n+1}|O_{1:n})dS_{n+1}$ is the normalization constant of the posterior $p(S_{n+1}|O_{1:n+1})$. In this way, the update step (7) incorporates the observation information, which is contained in the likelihood function, into the prior pdf through Bayes' formula and the procedure can be implemented recursively using (6) and (7).

If F and G are both linear and if $\{\gamma_n\}_{n\geq 0}$ and $\{\zeta_n\}_{n\geq 1}$ are Gaussian random variables, (6) and (7) can be calculated analytically by using Kalman formula and this gives Kalman filter [7]. The ensemble Kalman filter [8] uses Monte Carlo simulations of (5a) to obtain the covariance matrix by an ensemble of particles and then use the Kalman filter formalism for the update. For general nonlinear, non-Gaussian situations, (6) and (7) involve high dimensional integration and direct calculation of these formulas is analytically and computably prohibitive. One can approximate the recursive procedure using particle filters. The main theme of the particle filter method is to use the recursive formulas (6) and (7) to generate a cloud of particles whose empirical distribution follows the posterior pdf, i.e. $p(S_{n+1}|O_{1:n+1})$. In what follows, we use a bench mark particle filter method, the bootstrap filter algorithm [11], as our paradigm and briefly discuss the general framework of the particle filter method.

Assume at time instant n, we have a set of M particles, denoted by $\{s_n^{(m)}\}_{m=1}^M$, distributed according to an empirical distribution $\tilde{p}(S_n|O_{1:n})$ that approximates the conditional pdf $p(S_n|O_{1:n})$, where the empirical distribution $\tilde{p}(S_n|O_{1:n})$ is defined by

$$\tilde{p}(S_n|O_{1:n}) := \frac{1}{M}\sum_{m=1}^{M} \delta_{s_n^{(m)}}(S_n), \tag{8}$$

where δ_x is the Dirac delta function at x. In the prediction step of the particle filter method, we sample each $\tilde{s}_{n+1}^m$ from $\tilde{p}(S_n|O_{1:n})p(S_{n+1}|S_n)$ using (5a) to generate a new set of particles, denoted by $\{\tilde{s}_{n+1}^{(m)}\}_{m=1}^M$, that are distributed approximately according to $p(S_{n+1}|O_{1:n})$. The derived empirical distribution

$$\tilde{\pi}(S_{n+1}|O_{1:n}) := \frac{1}{M}\sum_{m=1}^{M} \delta_{\tilde{s}_{n+1}^{(m)}}(S_{n+1}) \tag{9}$$

is an approximation for $p(S_{n+1}|O_{1:n})$. After receiving the new observational measurement O_{n+1}, we carry out the Bayesian update and use the empirical approximation $\tilde{\pi}(S_{n+1}|O_{1:n})$ to replace $p(S_{n+1}|O_{1:n})$ in (7). As a result, we get a Monte Carlo approximation as following

$$\begin{aligned}\tilde{\pi}(S_{n+1}|O_{1:n+1}) :=& \frac{p(O_{n+1}|S_{n+1})\tilde{\pi}(S_{n+1}|O_{1:n})}{\int p(O_{n+1}|S_{n+1})\tilde{\pi}(S_{n+1}|O_{1:n})dS_{n+1}} \\ =& \frac{\sum_{m=1}^M p\left(O_{n+1}|\tilde{s}_{n+1}^{(m)}\right)\delta_{\tilde{s}_{n+1}^{(m)}}(S_{n+1})}{\sum_{m=1}^M p\left(O_{n+1}|\tilde{s}_{n+1}^{(m)}\right)},\end{aligned} \tag{10}$$

where $\tilde{\pi}(S_{n+1}|O_{1:n+1})$ is a weighted empirical distribution for the posterior $p(S_{n+1}|O_{1:n+1})$, which is also written by

$$\tilde{\pi}(S_{n+1}|O_{1:n+1}) = \sum_{m=1}^{M} W_{n+1}^{(m)} \delta_{\tilde{s}_{n+1}^{(m)}}(S_{n+1}),$$

where $W_{n+1}^{(m)} \propto p\left(O_{n+1}|\tilde{s}_{n+1}^{(m)}\right)$ is the importance weight with the property $\sum_{m=1}^M W_{n+1}^{(m)} = 1$.

A major problem of the aforementioned particle filter is that after many time steps lots of weights are negligible, and only a few particles have very large weights which reduces the effective sample size. This is often called the degeneracy problem [11,13]. The strategy to address the degeneracy problem is to introduce a *resampling step*, in which we produce more copies of particles with high weights and discard the rest. In the bootstrap particle filter, we use importance sampling to generate M samples, denoted by $\{s_{n+1}^{(m)}\}_{m=1}^{M}$, from the weighted empirical distribution $\tilde{\pi}(S_{n+1}|O_{1:n+1})$. In this way, we obtain an "unweighted" empirical distribution as follows

$$\tilde{p}(S_{n+1}|O_{1:n+1}) = \frac{1}{M}\sum_{m=1}^{M}\delta_{s_{n+1}^{(m)}}(S_{n+1}),$$

and $\tilde{p}(S_{n+1}|O_{1:n+1})$ would be the initial empirical distribution for the recursive step $n+1 \to n+2$.

The above "prediction-update-resample" procedure in the particle filter algorithm would provide an effective algorithmic framework to make estimation based on a dynamic model and observational data. In what follows, we introduce a common used nonlinear filter methodology to solve the parameter estimation problem.

3.2. Augmented filter for parameter estimation

Recall the state-space model for the parameter estimation problem (1)-(2), we want to get an estimation for the parameter θ in the state model (1) based on the dynamic function h and the observational data. In this subsection, we briefly discuss a popular on-line Bayesian parameter estimation method that estimates the target parameter by solving a corresponding nonlinear filtering problem [2,3,22]. To proceed, we consider the parameter θ as a sequence $\{\theta_n\}_{n\geq 0}$ and combine the parameter sequence with the state dynamics as a state process in the nonlinear filtering problem, i.e. $S_n := (\theta_n, X_n)$. Then, in the state process (5a), we define the function F as

$$F(S_n) = \begin{pmatrix} \theta_n \\ h(X_n, \theta_n) \end{pmatrix}.$$

In this way, we obtain a joint posterior distribution $p(S_{n+1}|O_{1:n+1}) = p(\theta_{n+1}, X_{n+1}|O_{1:n+1})$ based on some effective nonlinear filters, which would give us an estimation for the parameter sequence $\{\theta_n\}_{n\geq 0}$. Since θ is a static parameter, each θ_n in the sequence $\{\theta_n\}_{n\geq 0}$ will be some estimation for the true parameter θ based on the observational data. The methodology of implementing a nonlinear filtering algorithm to solve a combined state (the combination of the parameter with the state model) is typically called the "*augmented filter*" for the parameter estimation problem. Well known nonlinear filter implementations include the augmented particle filter and the augmented ensemble Kalman filter. However, after a more careful look at this augmented filter framework, the extended state process $\{S_n\}_{n\geq 0}$ does not have the forgetting property of a Markov process for the parameter part. Therefore, it's important to point out that the augmented particle filter algorithm will certainly suffer very severe degeneracy problem. The particles based algorithms could only search very limited region in the parameter space around the initial guess of θ and the resampling step in the bootstrap particle filter method would only reduce the diversity of the samples in the particle cloud – the weights $\{W_{n+1}^{(m)}\}_{m=1}^{M}$ tend to concentrate on only one sample very fast. To address the degeneracy problem, in the augmented particle filter, people typically consider the noise ω_n in the state process as $d+r$ dimensional Gaussian random variable [3]. The extra r dimensional noise is for the parameter sequence $\{\theta_n\}_{n\geq 0}$, which is given artificially by users. The artificial noise in the state process inflates diversity of the samples in the empirical distribution and would certainly rejuvenate the particle cloud. Therefore, standard particle filter methods can be applied to approximate $p(\theta_{n+1}, X_{n+1}|O_{1:n+1})$ and the standard analysis for the particle filters [23] would also be applicable, which demonstrates the convergence of the augmented particle filter for parameter estimation problem.

3.3. Parameter estimation through direct filter

In what follows, we will introduce the methodology of our novel nonlinear filter method for the parameter estimation problem as an alternative approach to the augmented filter that we discussed in Section 3.2. The central idea of this algorithm is similar to the augmented filter method due to the fact that it also uses the sequential Bayesian inference to estimate the parameter θ under the state-space model. However, different from the augmented filter method, in our novel approach we consider the parameter θ as the only state that we want to estimate and we construct a filtering algorithm which only provides inference estimation for the parameter itself. Since our approach focuses on using the nonlinear filter to estimate the parameter directly, we call this approach a direct filter algorithm for parameter estimation, or simply the "*direct filter*" for the convenience of presentation. To setup a state-space model for the parameter, we first model the parameter θ as a sequence with zero dynamics, i.e. define

$$\theta_{n+1} = \theta_n + \epsilon_n, \tag{11}$$

where ϵ_n is an artificial dynamic noise and θ_0 is the initial guess of the parameter in the dynamic model (1). In order to incorporate the measurement data, we use the observation $\{Y_n\}_{n\geq 1}$ in the original parameter estimation problem (2) to

be the observation process in the nonlinear filtering problem. Combining the parameter process (11) and the observation process (2), we obtain a state-space model for θ and can use some filtering algorithm to obtain the estimate based on the observation information $\mathcal{Y}_{1:n}$, where $\mathcal{Y}_{i:j} := \sigma(Y_{i:j})$ denotes the σ-algebra generated by the set of random variables $\{Y_i, Y_{i+1}, \ldots, Y_j\}$.

An immediate problem that we encounter from the above description of nonlinear filtering problem is that the parameter process $\{\theta_n\}_{n\geq 0}$ is an artificially defined pseudo-state process and the observation process $\{Y_n\}_{n\geq 1}$ does not provide measurements for θ_n directly. The bridge that connects the parameter and the observation is the dynamic model h, which is introduced in (1). To make the measurement data Y_n an effective observation for θ, we need to incorporate the model information in the algorithm. As a novelty in the direct filter methodology, we composite the dynamic model h into the observation function (2). Specifically, we use the dynamical system to represent X_{n+1} in the observational measurement, i.e. let $X_{n+1} = h(X_n, \theta) + w_n$ in the observation, and replace θ by the propagated parameter process θ_{n+1}. Therefore, the observational measurement is now described by

$$Y_{n+1} = H\Big(h(X_n, \theta_{n+1}) + w_n\Big) + \xi_{n+1}, \tag{12}$$

where H, h, X_n, w_n and ξ_{n+1} are all defined in the parameter estimation problem (1)-(2). We can see that the observation Y_{n+1} defined in (12) is still the original observation in the parameter estimation problem and we only provide an equivalent alternative expression for the observation function. Apparently, the observation (12) is an indirect observation for the parameter θ_{n+1} with a nonlinear observation function $Hh(X_n, \theta_{n+1})$. Here X_n could also be described as a functional of θ determined by the dynamic model h and the model noise w_n. A new observational noise for the parameter θ_{n+1} will be the model noise w_n combined with the observational noise ξ_{n+1}. We define the combined observational noise by

$$\eta_{n+1} := Hw_n + \xi_{n+1} \tag{13}$$

which is still a Gaussian noise since both w_n and ξ_{n+1} are Gaussian random variables, and we denote the invariant variance of η by R. In this way, we incorporate the dynamic model structure in the algorithm and preserve the relation between the parameter and the model in the direct filter method. As a result, we introduce the following nonlinear filtering problem

$$\theta_{n+1} = \theta_n + \epsilon_n, \tag{14a}$$

$$Y_{n+1} = Hh\Big(X_n(X_{n-1}, \theta_{n+1}, w_{n-1}), \theta_{n+1}\Big) + \eta_{n+1}, \tag{14b}$$

where $\{\theta_n\}_{n\geq 0}$ is the parameter state process, $\{Y_n\}_{n\geq 1}$ is the observation process, and we write X_n as a functional of X_{n-1}, θ_{n+1} and w_{n-1}. The goal of the nonlinear filtering problem (14) is to get the posterior density $p(\theta_n|\mathcal{Y}_{1:n})$ for the parameter θ_n given the observation information $\mathcal{Y}_{1:n}$ at each step n. Even though the algorithm is similar to the augmented filter approach, the direct filter method is essentially different. The major difference is that the dynamic model is incorporated in the observation function in the direct filter method, not in the state process as in the augmented filter method.

In order to solve the state-space model (14) corresponding to the parameter estimation problem (1)-(2), we adopt the recursive procedure introduced in Section 3.1. To incorporate the observational data Y, we apply the Bayes' formula with likelihood function $p\big(Y_{n+1}|\theta\big)$, where

$$\begin{aligned} p\big(Y_{n+1}|\theta_{n+1}\big) &= \int p\big(Y_{n+1}|X_{n+1}\big)p\big(X_{n+1}|\theta_{n+1}\big)dX_{n+1} \\ &= \int \left(\int p\big(Y_{n+1}|X_{n+1}\big)p\big(X_{n+1}|X_n, \theta_{n+1}\big)p\big(X_n|\theta_{n+1}\big)dX_n\right)dX_{n+1}. \end{aligned}$$

Since we do not have the "exact" state of the dynamical system X_n available, the above likelihood function is not computable directly. One solution to estimate the dynamical system X_n is to sample it using the dynamic model and approximate the integral as in [15]. However, this procedure is based on sequential sampling and typically suffers from sampling the long trajectory of $X_{1:n}$ with the unknown parameter θ. To avoid this difficulty, instead of using a sequential sampling method to derive an approximation for the state X_n, the direct filter uses the observational data Y_n to approximate X_n. This can be derived from the linear relation between Y_n and X_n in the observation, i.e. $Y_n = HX_n + \xi_n$ and we obtain $X_n = H^{-1}(Y_n - \xi_n)$ with the assumption of the invertibility of H. Then we drop the observational noise ξ_n and get the direct approximation $X_n(Y_n) \approx H^{-1}Y_n$. In this way, the above likelihood function $p\big(Y_{n+1}|\theta_{n+1}\big)$ is approximated as

$$p\big(Y_{n+1}|\theta_{n+1}\big) \approx \int p\big(Y_{n+1}|X_{n+1}\big)p\big(X_{n+1}|H^{-1}Y_n, \theta_{n+1}\big)dX_{n+1}. \tag{15}$$

It's worthy to mention that since we use the current observation Y_n to describe X_n, the error $H^{-1}\xi_n$ would not be accumulated through the parameter estimation process. As a result of the direct approximation for X_n, we can substitute the state X_n in the observation function in (14b) and introduce an approximate state space model for θ as

$$\theta_{n+1} = \theta_n + \epsilon_n, \tag{16a}$$

$$\tilde{Y}_{n+1} = Hh(H^{-1}\tilde{Y}_n, \theta_{n+1}) + \eta_{n+1}, \tag{16b}$$

where $\tilde{Y}_n = Y_n$ is the original observational measurement for X_n.

We can see that the nonlinear filtering problem (16) is an approximation for the nonlinear filtering problem (14) by dropping the error $H^{-1}\xi_n$ when approximating X_n, which results in the approximate likelihood function (15). It's important to point out that, obtaining the best approximation for the parameter θ_{n+1} in (16) based on the observation information $\tilde{\mathcal{Y}}_{1:n+1}$, i.e. $E[\theta_{n+1}|\tilde{\mathcal{Y}}_{1:n+1}]$, achieves the same goal as in the nonlinear filtering problem (14), which aims to get the best approximation for the parameter θ_{n+1} based on the observation information $\mathcal{Y}_{1:n+1}$, i.e. $E[\theta_{n+1}|\mathcal{Y}_{1:n+1}]$. To verify this statement, we compare the random variables that generate σ-algebras $\mathcal{Y}_{1:n+1}$ and $\tilde{\mathcal{Y}}_{1:n+1}$. To proceed, we first consider the original observational measurement Y_{n+1}, which is now in the form of (14b). The σ-algebra of Y_{n+1}, $\sigma(Y_{n+1})$, is generated by X_n, θ_{n+1}, w_n and ξ_{n+1}. Since θ_{n+1} is adaptive to $\sigma(\epsilon_{0:n})$ and X_n is adaptive to $\sigma(w_{0:n-1})$ and $\sigma(\theta_{n+1})$, we have $\sigma(\theta_{n+1}) = \sigma(\epsilon_{0:n})$ and $\sigma(X_n) = \sigma(w_{0:n-1}, \epsilon_{0:n})$. Therefore, $\sigma(Y_{n+1})$ is also a σ-algebra generated by $w_{0:n}$, $\epsilon_{0:n}$ and ξ_{n+1}, or we can write $\sigma(Y_{n+1}) = \sigma(w_{0:n}, \epsilon_{0:n}, \xi_{n+1})$. On the other hand, for the modified observation $\tilde{Y}_{n+1}$, the σ-algebra of $\tilde{Y}_{n+1}$, $\sigma(\tilde{Y}_{n+1})$, is generated by $\tilde{Y}_n$, θ_{n+1}, w_n and ξ_{n+1}. It's easy to see that $\sigma(\tilde{Y}_n) = \sigma(w_{0:n-1}, \epsilon_{0:n-1}, \xi_n)$, which gives $\sigma(\tilde{Y}_{n+1}) = \sigma(w_{0:n}, \epsilon_{0:n}, \xi_n, \xi_{n+1})$. By comparing the random variables that generate $\sigma(Y_{n+1})$ and $\sigma(\tilde{Y}_{n+1})$, we get $\mathcal{Y}_{1:n+1} = \tilde{\mathcal{Y}}_{1:n+1}$ given $\xi_0 = 0$, which means $\tilde{\mathcal{Y}}_{1:n+1}$ contains the same amount of information as $\mathcal{Y}_{1:n+1}$. In this way, although the observational noise is dropped when approximating X_n (by using $H^{-1}Y_n$ in (16b)), the information about the observational noise is not ignored since the filtering problem (16) in the direct filter is unbiased and would also provide $E[\theta_{n+1}|\mathcal{Y}_{1:n+1}]$ as the estimation for the parameter.

Remark 3.1. The linear observation assumption in (2) can be broadened to invertible observation functions. An invertible observation function $H(\cdot)$ that provides a representation for X_n by using Y_n could be used to design the modified observation process in the direct filter. In large amount of application problems, the observations are invertible.

From the above discussion, we can see that the direct filter method lies in the general framework of online Bayesian parameter estimation and it is based on solving nonlinear filtering problems. Although the direct filter method looks similar to the augmented filter approach, there are two major advantages:

- In the direct filter, we don't need to use the estimations on the model state to drive the estimations on the parameter. In this way, the dimension of the direct filtering problem (16) is the same as the dimension of the parameter θ. This can be considered as a dimensionality reduction strategy and would address the "curse of dimensionality" when solving nonlinear filtering problems in online parameter estimation. As a result, the direct filter could solve the parameter estimation problem more efficiently. This is especially more efficient when the dynamical system (1) is a very high dimensional model and the parameter θ is a relatively low dimensional vector.
- In the direct filter, the observational information about the dynamical system is used directly to estimate the parameter as the update step, which incorporates the data with a likelihood function. Moreover, the direct filter can be considered as a method that projects the observational information, as well as the model information, to the parameter space through the modified observation process defined in (16b) and the Bayesian inference. This allows us to focus on the connection between the observation and the parameter, and would avoid unnecessary distractions in state estimation. For some complicated models, tracking the state variable is already very challenging and might dominate the parameter estimation task which we are more interested in this work. As a result, although the direct filter does not provide estimations on the model state, it could estimate parameters more accurately.

3.4. Particle filter implementation of the direct filter

In this subsection we provide a specific particle filter algorithm to solve the nonlinear filtering problem (16). Any other nonlinear filter methods would also be applied for (16) as well.

At time instant n, assume that we have a set of M particles, denoted by $\{\zeta_n^{(m)}\}_{m=1}^M$, as an empirical distribution for the parameter θ_n given $\tilde{Y}_{1:n}$ that describes the conditional pdf $p(\theta_n|\tilde{Y}_{1:n})$, and we define the empirical distribution $\tilde{p}(\theta_n|\tilde{Y}_{1:n})$ by

$$\tilde{p}(\theta_n|\tilde{Y}_{1:n}) := \frac{1}{M}\sum_{m=1}^{M} \delta_{\zeta_n^{(m)}}(\theta_n).$$

In the prediction step of the direct filter, we essentially inflate some artificial noise to the parameter particles cloud as we use zero dynamics in the state process (16a). For each particle $\zeta_n^{(m)}$, we derive the corresponding particle $\tilde{\zeta}_{n+1}^{(m)}$ by adding a version of noise $\epsilon_n^{(m)}$, i.e. $\tilde{\zeta}_{n+1}^{(m)} = \zeta_n^{(m)} + \epsilon_n^{(m)}$, and derive the empirical distribution that approximates $p(\theta_{n+1}|\tilde{Y}_{1:n})$ as

$$\tilde{\pi}(\theta_{n+1}|\tilde{Y}_{1:n}) := \frac{1}{M}\sum_{m=1}^{M} \delta_{\tilde{\zeta}_{n+1}^{(m)}}(\theta_{n+1}). \tag{17}$$

When the observation $\tilde{Y}_{n+1}$ is available, we update the prior distribution $\tilde{\pi}(\theta_{n+1}|\tilde{Y}_{1:n})$. Although the observation $\tilde{Y}_{n+1}$ defined in (16b) is different from the observation Y_{n+1}, we still apply the Bayesian formula to the predicted distribution and get a Monte Carlo approximation for the posterior distribution as

$$\tilde{\pi}(\theta_{n+1}|\tilde{Y}_{1:n+1}) := \frac{\sum_{m=1}^{M} p\big(\tilde{Y}_{n+1}\big|\tilde{\zeta}_{n+1}^{(m)}\big)\delta_{\tilde{\zeta}_{n+1}^{(m)}}(\theta_{n+1})}{\sum_{m=1}^{M} p\big(\theta_{n+1}\big|\tilde{\zeta}_{n+1}^{(m)}\big)}, \tag{18}$$

where $\tilde{\pi}(\theta_{n+1}|\tilde{Y}_{1:n+1})$ is a weighted empirical posterior distribution that approximates the posterior distribution $p(\theta_{n+1}|\tilde{Y}_{1:n+1})$. Here, $p\big(\tilde{Y}_{n+1}\big|\tilde{\zeta}_{n+1}^{(m)}\big)$ is the likelihood function defined by

$$p\big(\tilde{Y}_{n+1}\big|\tilde{\zeta}_{n+1}^{(m)}\big) := \exp\Big(-\frac{1}{2}\|Hh(H^{-1}\tilde{Y}_n, \tilde{\zeta}_{n+1}^{(m)}) - \tilde{Y}_{n+1}\|_R^2\Big),$$

where the norm $\|\cdot\|_R$ is defined by $\|\alpha\| := \alpha R^{-1}\alpha$ and R is the invariant variance of the observational noise η.

To address the degeneracy problem in the particle filter method, we also resample the particles by using the importance sampling method and generate M samples $\{\zeta_{n+1}^{(m)}\}_{m=1}^{M}$ from $\tilde{\pi}(\theta_{n+1}|\tilde{Y}_{1:n+1})$. We denote the resulted unweighted empirical distribution by

$$\tilde{p}(\theta_{n+1}\big|\tilde{Y}_{1:n+1}) = \frac{1}{M}\sum_{m=1}^{M}\delta_{\zeta_{n+1}^{(m)}}(\theta_{n+1}),$$

and use $\tilde{p}(\theta_{n+1}\big|\tilde{Y}_{1:n+1})$ to initialize our filtering algorithm for the next step.

At every time step $n \geq 1$, when we obtain the approximate pdf $\tilde{p}(\theta_n\big|\tilde{Y}_{1:n})$, we calculate the corresponding conditional expectation $\tilde{E}[\theta_n\big|\tilde{\mathcal{Y}}_{1:n}]$ as our estimate parameter at time step n and denote it by $\tilde{\theta}_n := \tilde{E}[\theta_n\big|\tilde{\mathcal{Y}}_{1:n}]$. Since the artificial parameter noise ϵ_n in the direct filter method is an invariant Gaussian noise with constant covariance and the observational noise ξ_n might have strong influence to a certain estimation, when we get $\tilde{E}[\theta_n\big|\tilde{\mathcal{Y}}_{1:n}]$ as the estimate parameter derived from the nonlinear filtering problem (16), we expect that the single conditional expectation at time instant n may not be a very good estimation for the parameter θ. To balance the influence of noises, in practice, we let $\tilde{\theta} := \frac{1}{n-l}\sum_{i=l}^{n}\tilde{E}[\theta_i\big|\tilde{\mathcal{Y}}_{1:i}]$ be our estimation for the parameter θ in the direct filter method, where $l \geq 1$ is a user defined burn-in step number.

In what follows, we summarize our direct filter method with particle filter implementation, and describe how we obtain a sequence of dynamical estimation for the parameter $\{\tilde{\theta}_n\}_{n\geq 0}$, and the invariant estimation $\tilde{\theta}$.

Initialization: Generate a set of particles $\{\zeta_0^{(m)}\}_{m=1}^{M}$ that reflect the initial guess for the parameter θ. Input the artificial dynamic noise level ϵ and the user defined burn-in step number l.

Particle filter implementation for *direct filter*: For $n = 0, 1, 2, \cdots$,

- **Prediction.** Propagate parameter particles by adding noises $\{\epsilon_n^{(m)}\}_{m=1}^{M}$ to get $\{\tilde{\zeta}_{n+1}^{(m)}\}_{m=1}^{M}$. The empirical distribution that approximates the prior distribution $p(\theta_{n+1}|\tilde{Y}_{1:n})$ is $\tilde{\pi}(\theta_{n+1}|\tilde{Y}_{1:n}) = \frac{1}{M}\sum_{m=1}^{M}\delta_{\tilde{\zeta}_{n+1}^{(m)}}(\theta_{n+1})$.
- **Update.** Upon receiving the new observational data Y_{n+1}, apply the Bayesian formula to get a Monte Carlo approximation for the posterior distribution as

$$\tilde{\pi}(\theta_{n+1}|\tilde{Y}_{1:n+1}) = \frac{\sum_{m=1}^{M} p\big(\tilde{Y}_{n+1}\big|\tilde{\zeta}_{n+1}^{(m)}\big)\delta_{\tilde{\zeta}_{n+1}^{(m)}}(\theta_{n+1})}{\sum_{m=1}^{M} p\big(\theta_{n+1}\big|\tilde{\zeta}_{n+1}^{(m)}\big)},$$

  where we have $\tilde{Y}_n = Y_n$ and the likelihood function is calculated by $p\big(\tilde{Y}_{n+1}\big|\tilde{\zeta}_{n+1}^{(m)}\big) = \exp\big(-\frac{1}{2}\|Hh(H^{-1}\tilde{Y}_n, \tilde{\zeta}_{n+1}^{(m)}) - \tilde{Y}_{n+1}\|_R^2\big)$.
- **Resample.** Resample particles with importance distribution $\tilde{\pi}(\theta_{n+1}|\tilde{Y}_{1:n+1})$ to get $\{\zeta_{n+1}^{(m)}\}_{m=1}^{M}$. The unweighted empirical posterior distribution $\tilde{p}(\theta_{n+1}\big|\tilde{Y}_{1:n+1}) = \frac{1}{M}\sum_{m=1}^{M}\delta_{\zeta_{n+1}^{(m)}}(\theta_{n+1})$ is the approximation for the conditional distribution for the parameter, i.e. $p(\theta_{n+1}\big|\tilde{Y}_{1:n+1})$. The estimation for the parameter at time step n is $\tilde{\theta}_n = \tilde{E}[\theta_n\big|\tilde{\mathcal{Y}}_{1:n}]$.

Calculate the invariant estimation: The invariant estimation for the parameter $\tilde{\theta}$ is calculated by $\tilde{\theta} = \frac{1}{n-l}\sum_{i=l}^{n}\tilde{E}[\theta_i\big|\tilde{\mathcal{Y}}_{1:i}]$ with distributions $\{\tilde{p}(\theta_i\big|\tilde{\mathcal{Y}}_{1:i})\}_{i=l}^{n}$.

4. Numerical experiments

In this section, we examine the performance of our direct filter method for the parameter estimation problem through three numerical examples. The first example is a simple model problem. We estimate four parameters that govern a two dimensional stochastic dynamics. In the second example, we estimate the parameter in a high dimensional chaotic dynamical

system – a noise perturbed 10 dimensional Lorenz 96 model governed by a 1 dimensional parameter. The third example is a scientific parameter estimation problem that we encountered in practical material science applications [24]. In this problem, we want to make estimations for the parameters that determine energy potential models on material surface. The measurements we receive are atom positions observed from electron microscopy. To demonstrate the accuracy of our algorithm, we compare our method with the augmented filter implemented by the particle filter and the Ensemble Kalman filter [8,25,26], which is a popular online parameter estimation methods. In the particle filter algorithm, we use sequential sampling with importance sampling (SIR). More sophisticated particle filter methods could be used to improve the performance, but won't change the nature of methodology.

Example 1

We consider the following dynamical system

$$\begin{aligned} dX_t^1 =& \Big(a_1 \sin(X_t^2) + a_2 \frac{X_t^1}{1+|X_t^1|}\Big)dt + \sigma^1 dW_t^1, \\ dX_t^2 =& \Big(a_3 \cos(X_t^1) + a_4 \frac{X_t^2}{1+|X_t^2|}\Big)dt + \sigma^2 dW_t^2, \end{aligned} \tag{19}$$

where $\boldsymbol{X}_t := (X_t^1, X_t^2)$ describes the state of the system, (W_t^1, W_t^2) is a standard two dimensional Brownian motion and $\int \cdot dW$ is Itô integral as defined in [27]. The parameter vector θ in the above dynamical system is composed by four components, i.e. $\theta = (a_1, a_2, a_3, a_4)$, and the dynamics is in the form of a 2 dimensional stochastic differential equation (SDE) system. To implement our algorithm, we discretize the SDE system (19) with Euler-Maruyama scheme and write it as a sequence

$$\begin{aligned} X_{n+1}^1 =& X_n^1 + \Big(a_1 \sin(X_n^2) + a_2 \frac{X_n^1}{1+|X_n^1|}\Big)\Delta t + \sigma^1 w_n^1, \\ X_{n+1}^2 =& X_n^2 + \Big(a_3 \cos(X_n^1) + a_4 \frac{X_n^2}{1+|X_n^2|}\Big)\Delta t + \sigma^2 w_n^2, \end{aligned} \tag{20}$$

where w_n^1 and w_n^2 are two Gaussian random variables with variance Δt. To estimate the parameters a_1, a_2, a_3 and a_4, we receive an observation Y_{n+1} with linear dependence on X_{n+1} as follows

$$\boldsymbol{Y}_{n+1} = H\boldsymbol{X}_{n+1} + \boldsymbol{\xi}_{n+1},$$

where ξ_{n+1} is a standard 2 dimensional Gaussian random variable with variance 0.1 and H is a 2×2 constant matrix. In this example, we observe the dynamics $\boldsymbol{X}_t$ for 200 times with 0.05 second observational gap, i.e. $\Delta t = 0.05$. To proceed, we choose the parameters to be $a_1 = 4$, $a_2 = 2$, $a_3 = 3$, $a_4 = 5$, and let $\sigma^1 = \sigma^2 = 0.1$, $H = \begin{pmatrix} 5 & 0 \\ 0 & 3 \end{pmatrix}$. We assume that the initial guess for the parameter follows a 4 dimensional normal distribution with mean $\theta_0 := (1, 1, 1, 1)$ and variance 1 in each dimension.

In Fig. 1, we plot a sample trajectory of the dynamical system to describe the state behavior that is governed by the parameter. In Fig. 2, we present the estimated parameters a_1 - a_4 by using the direct filter over the receptions of measurements Y_n, in comparison with the augmented particle filter (AugPF) and the augmented ensemble Kalman filter (EnKF). The black line in each subplot shows the invariant true parameter for a_1 - a_4 respectively. The blue curves marked by circles give our estimated parameters obtained by using the direct filter; the red curves marked by triangles are estimated parameters obtained by using the AugPF; and the green curves marked by crosses show the estimated parameters obtained by using the EnKF. For both the direct filter and the AugPF, we use 2000 particles to represent the conditional distribution in the particle filter algorithms. For the EnKF, we use 20 ensembles, which is sufficient for this relatively low dimensional problem. We use the solution of the corresponding nonlinear filtering problem, i.e. $\tilde{E}[\theta_n | \tilde{\mathcal{Y}}_{1:n}]$, as our estimated parameters at time step n. From this figure, we can see that after a few burn-in steps, the direct filter could capture the true parameters quite accurately, while it's rather difficult for the AugPF to get good estimations for any of these parameters. On the other hand, the EnKF could capture parameters a_1 and a_2 well, but it performs poorly in estimating a_3 and a_4.

For this example, the AugPF basically deals with a 6 dimensional nonlinear filtering problem, which is a moderately high dimensional problem with high nonlinearity. Although the AugPF does not give reliable estimates with 2000 particles, we want to point out that the AugPF would achieve better performance with more particles involved – since typically the particle filter method would work better for higher dimensional problems with more particles. To verify this statement, we repeat the above experiment with 30,000 particles for the AugPF and show the parameter estimation performance in Fig. 3. From this figure, we can see that the performance of the AugPF is much improved as we are using more particles in the algorithm. Apparently, although we only used 2000 particles in our direct filter method, it still outperforms the AugPF with 30,000 particles. This demonstrates both the efficiency advantage and the accuracy advantage of the direct filter.

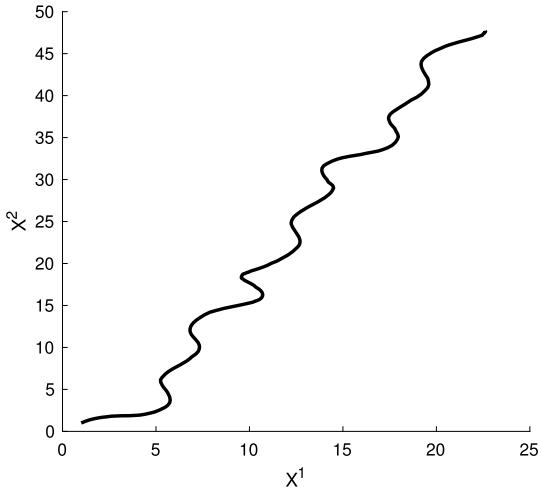
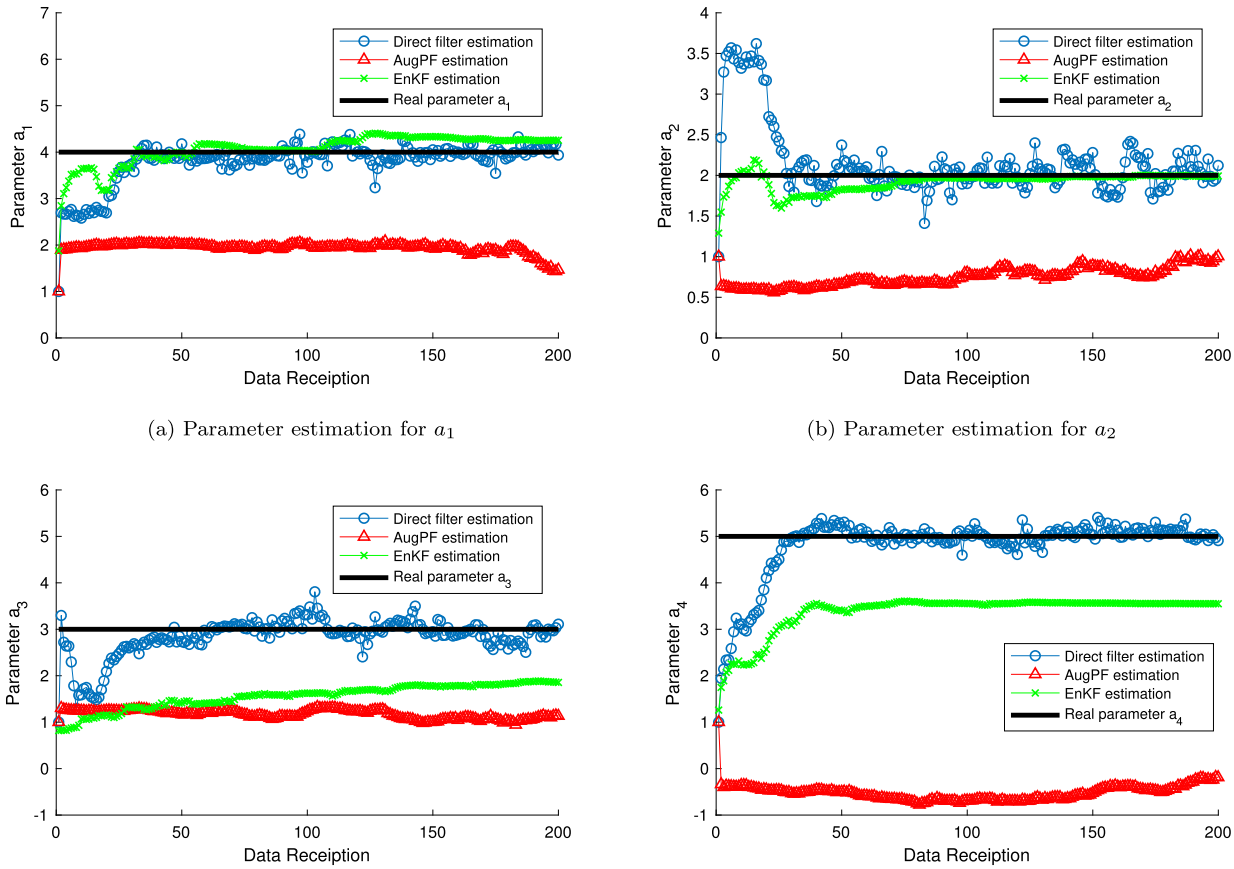


Fig. 1. A sample trajectory of the dynamical model on the time interval [0, 10].



(a) Parameter estimation for a_1

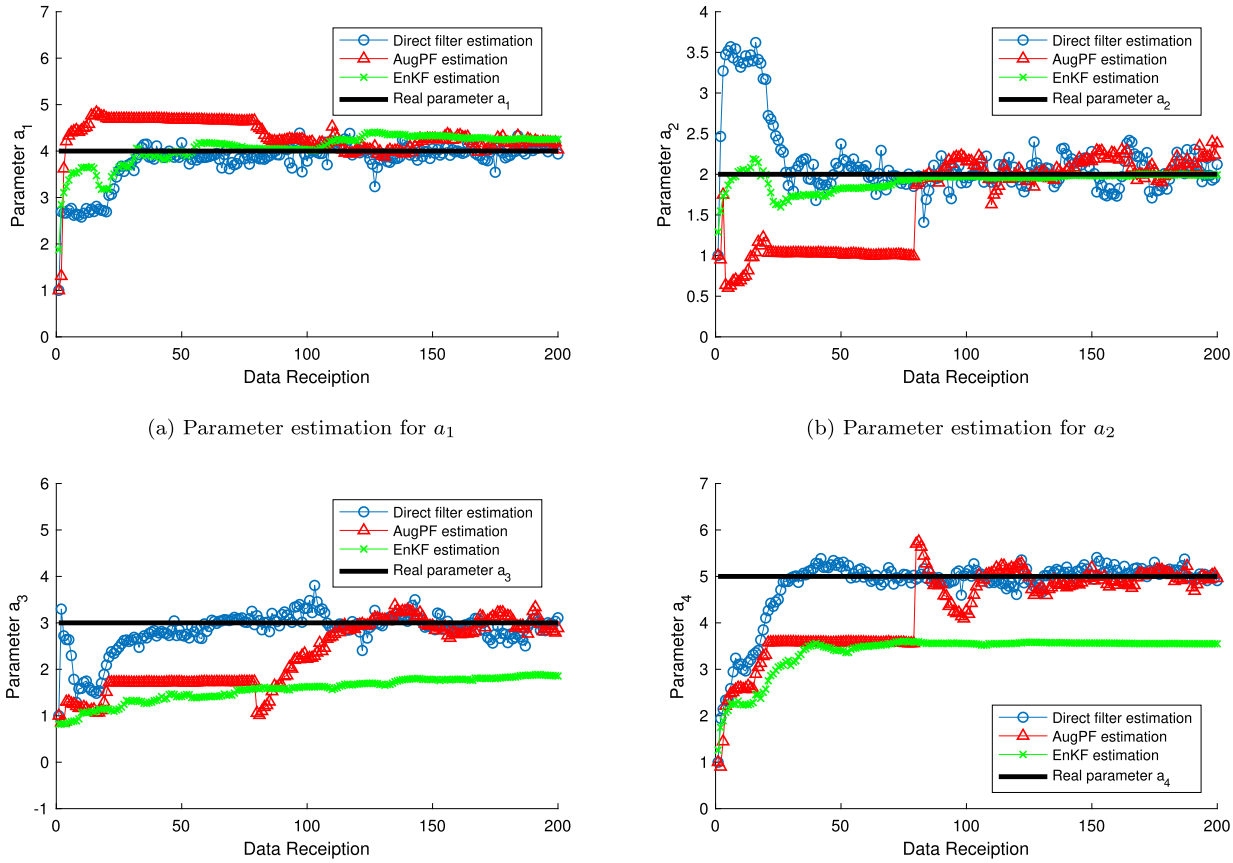
(b) Parameter estimation for a_2

(c) Parameter estimation for a_3

(d) Parameter estimation for a_4

Fig. 2. Comparison of parameter estimation between the direct filter, the AugPF and the EnKF with 2000 particles for the direct filter and the AugPF, and 20 ensembles for the EnKF. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

In order to further examine the performance of our algorithm, we repeat the above parameter estimation experiment 100 times using different realizations of random samples and show the root mean square errors (RMSEs) of each method in estimating the parameter. To demonstrate the robustness of the direct filter method, we allow the target parameter to be a random variable, which is designed to be the original "true" parameter with perturbation of some noises. Moreover,



(a) Parameter estimation for a_1

(b) Parameter estimation for a_2

(c) Parameter estimation for a_3

(d) Parameter estimation for a_4

Fig. 3. Comparison of parameter estimation between the direct filter, the AugPF and the EnKF with 2000 particles for the direct filter, 30,000 particles for the AugPF, and 20 ensembles for the EnKF.

we allow the observation matrix H to be randomly selected in each repeated test to show the stability of the direct filter in terms of selection of the observation matrix. In addition, we assume that the initial state $\boldsymbol{X}_0$ to be a random variable to bring in more uncertainties in this experiment. Specifically, in this experiment we let the parameter θ to be perturbed by a Gaussian noise with zero mean and 0.25 variance in each dimension, and the initial state $\boldsymbol{X}_0$ is perturbed by a Gaussian noise with zero mean and 0.01 variance in each dimension. For the observation matrix H, instead of adding a perturbation, we let $H = \begin{pmatrix} h_{11} & 0 \\ 0 & h_{22} \end{pmatrix}$, where h_{11} and h_{22} are integers randomly selected from 5 to 15 independently. In this way, for each realization we have different target parameter θ, different initial state $\boldsymbol{X}_0$, and different observation matrix H.

In Fig. 4, we show the RMSEs of the direct filter compared with the AugPF and the EnKF. We calculate the RMSEs at each time step n by

$$RMSEs(n) := \sqrt{\frac{1}{MC}\sum_{i=1}^{MC} \|\tilde{\theta}_n - \theta\|^2}, \tag{21}$$

where $\|\cdot\|$ is the Euclidean norm, $\tilde{\theta}_n$ is the estimated parameter at time step n, MC is the number of repeated tests, which is chosen to be $MC = 100$ in this experiment. From Fig. 4, we can see that the EnKF has the highest RMSEs among three methods, the AugPF with 30,000 particles performs better than the EnKF, and the direct filter with 2,000 particles has the lowest RMSEs, which is around 0.5 combining the errors of all 4 parameters. Since the RMSEs are calculated from 100 realizations, this experiment shows that the direct filter outperforms the AugPF and the EnKF in this example.

Example 2

In this example, we consider the Lorenz 96 model driven by a parameter α. For the d-th dimensional state X_t^d of the Lorenz 96 system, we have

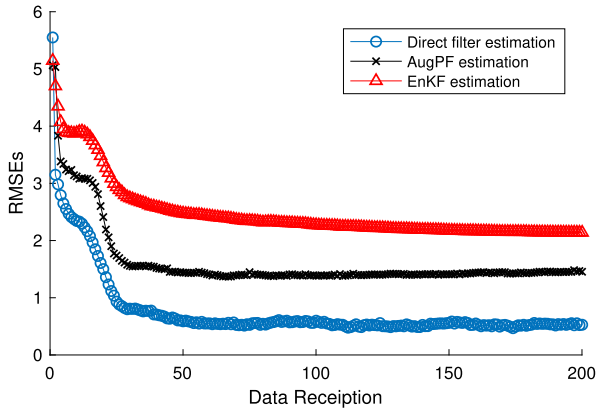


Fig. 4. Comparison of RMSEs between the direct filter, the AugPF and the EnKF.

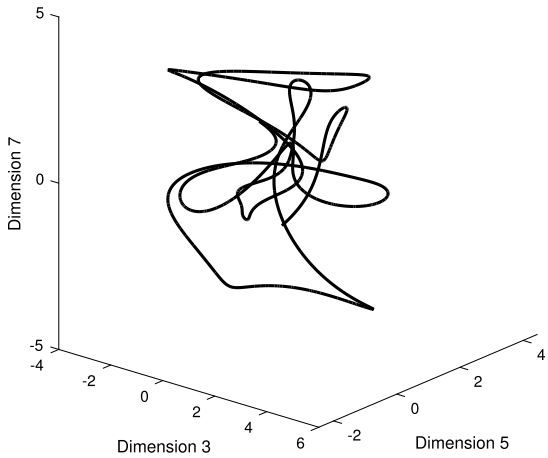


Fig. 5. A sample trajectory of Lorenz 96 model with respect to dimensions 3, 5 and 7.

$$dX_t^d = \Big(\alpha \cdot (X_t^{d+1} - X_t^{d-2})X_t^{d-1} - X_t^d + F\Big)dt + \sigma dW_t^d,\ d = 1, 2, \ldots, D, \tag{22}$$

where $\boldsymbol{X}_t := (X_t^1, X_t^2, \ldots, X_t^D)$ is the state of Lorenz 96 model, $D \in \mathbb{N}$ is the dimension of the system, F is a forcing term, and $\boldsymbol{W}_t := (W_t^1, W_t^2, \ldots, W_t^D)$ is a D dimensional standard Brownian motion. In the above dynamics, we let $X_t^{-1} = X_t^{D-1}$, $X_t^0 = X_t^D$, $X_t^{D+1} = X_t^1$, and we are interested in estimating the scaler parameter α in the model. It is well known that the Lorenz 96 model is a chaotic system, which is very difficult to track the state $\boldsymbol{X}_t$ effectively when the dimension D is getting large. In this example, we let $D = 10$, $F = 8$, $\sigma = 0.2$ and choose the true parameter to be $\alpha = 5$. In Fig. 5, we plot a sample trajectory of Lorenz 96 model and present the 3 dimensional trajectory of the $\boldsymbol{X}_t$ process in dimensions $d = 3$, $d = 5$ and $d = 7$. We can see from the figure that the trajectory has chaotic behavior and the state is difficult to predict. In the numerical experiment, we use noise perturbed direct observations of the state $\boldsymbol{X}_t$, i.e. H is the identity matrix $\boldsymbol{I}_D$, to estimate the parameter α and the observational noise is chosen to be a Gaussian random variable with variance 0.01. We also assume that the initial guess for the parameter follows a normal distribution with mean $\theta_0 := 30$ and variance 2 in each dimension.

To examine the effectiveness and accuracy of the direct filter, we estimate the parameter α by observing the dynamics $\boldsymbol{X}_t$ for 1 second and we receive observational measurements in every 0.01 second, i.e. $\Delta t = 0.01$. Also, we discretize the dynamical system (22) with Euler-Maruyama scheme. Since we only need to estimate a one-dimensional parameter as our target parameter in the direct filter method, we use 500 particles in the particle filter part of the direct filter algorithm. In this experiment, we compare our direct filter with the AugPF and the EnKF and demonstrate the performance of all three methods in parameter estimation. For the AugPF, we use 5,000 particles to represent the conditional distribution in the nonlinear filtering problem, which is 10 times of particles as we use in the direct filter. For the EnKF method, we use altogether 100 ensembles to estimate the augmented state, which is sufficient for this 11 dimensional problem. For each method, we also use the solution of the corresponding nonlinear filtering problem, i.e. $\tilde{E}[\theta_n | \tilde{\mathcal{Y}}_{1:n}]$, as our estimated parameter $\tilde{\theta}_n$ at time step n. In Fig. 6, we present the estimated parameter α with respect to the reception of observational data for X_t. The black solid line indicates the true parameter $\alpha = 5$; the blue curve marked by circles gives our estimated

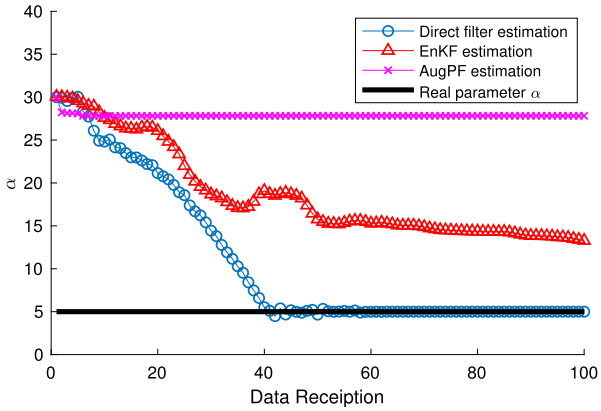
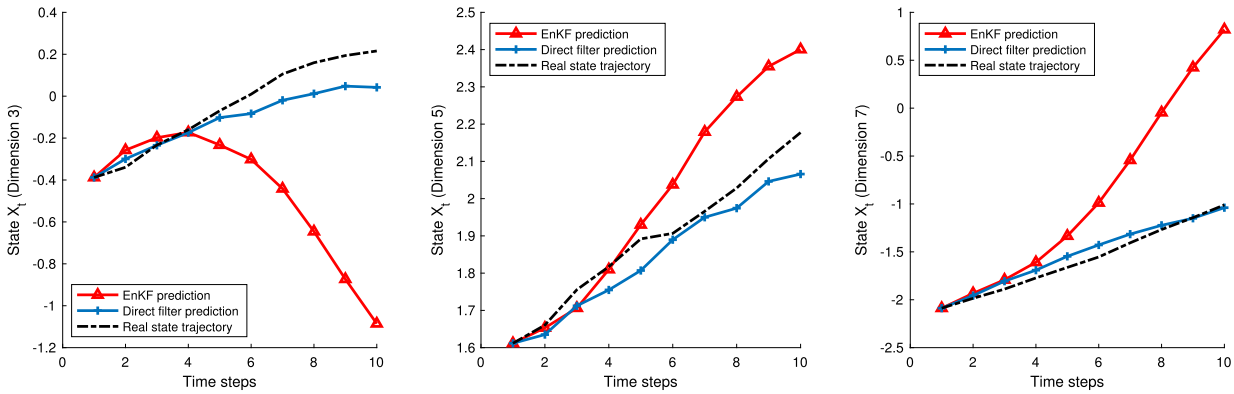


Fig. 6. Comparison of parameter estimation between the direct filter, AugPF and the EnKF with 500 particles for the direct filter, 5,000 particles for the AugPF, and 100 ensembles for the EnKF.



(a) Dimension 3 (b) Dimension 5 (c) Dimension 7

Fig. 7. Prediction comparison: (a) comparison in dimension 3; (b) comparison in dimension 5; comparison in dimension 7.

parameter obtained by using the direct filter; the magenta curve marked by crosses shows the estimated parameter obtained by using the AugPF; and the red curve marked by triangles is the estimated parameter obtained by using the EnKF method. We can see from the figure that the direct filter using only 500 particles approaches the true parameter after about 40 estimation steps. On the other hand, the EnKF does not converge to the true parameter in the entire observation period, and the AugPF does not provide useful convergence trend, even with 5000 particles, in this parameter estimation problem due to the high dimensionality of the state and the insufficient number of particles to carry out the particle filter algorithm. Actually, since the particle filter methods generally suffer from "curse of dimensionality", significant amount of particles need to be used to handle this 11 dimensional filtering problem, which is prohibitive in this experiment.

To demonstrate the influence of parameter estimation in this Lorenz 96 model, we compute the estimated parameters by the direct filter and the EnKF (since the AugPF does not provide good estimations in this example) and use these estimated parameters to predict the state of Lorenz system in a short period. To proceed, we use 50 burn-in steps, i.e. $l = 50$, and calculate the invariant mean parameter estimation in the direct filter as $\tilde{\alpha} = 5.0013$. Similarly, the EnKF estimation is $\hat{\alpha} = 14.5917$. In Fig. 7, we initialize the Lorenz 96 model with some random condition as state $\boldsymbol{X}_0$ and predict the state $\boldsymbol{X}_t$ in 10 simulation steps. The black dashed curve is the state of the Lorenz 96 model with true parameter α, the blue curve is the predict state of the model with estimated parameter obtained from the direct filter method, and the red curve is the predict state of the model obtained from the EnKF. We can see from the figure that for this short term prediction, the direct filter estimation gives good prediction to the original model while the EnKF gives totally different path due to the large error in the estimate parameter.

To further examine the performance of each method, we repeat the above parameter estimation experiment 100 times using different realizations of random samples and compute the RMSEs. In this experiment, we also let the target parameter to be the "true" parameter α perturbed by a Gaussian noise with zero mean and 0.25 variance, and the initial state of the Lorenz 96 model $\boldsymbol{X}_0$ is perturbed by a Gaussian noise with zero mean and 0.01 variance in each dimension. In Fig. 8, we plot the RMSEs, as defined in (21), of the direct filter, the AugPF and the EnKF in estimating the parameter. From this figure, we can see that the AugPF does not converge to the true parameter due to the high dimensionality, the EnKF shows the

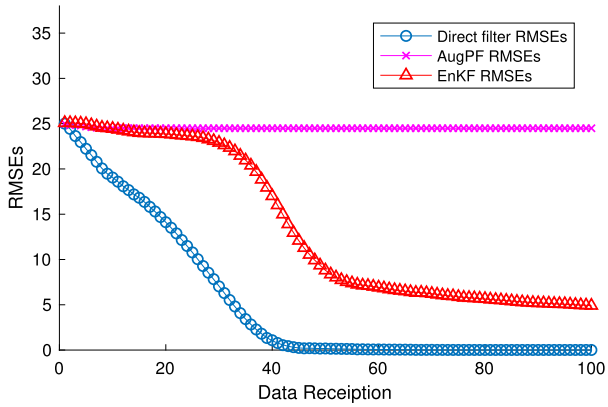


Fig. 8. Comparison of RMSEs between the direct filter, the AugPF and the EnKF with 500 particles for the direct filter, 5,000 particles for the AugPF and 100 ensembles for the EnKF.

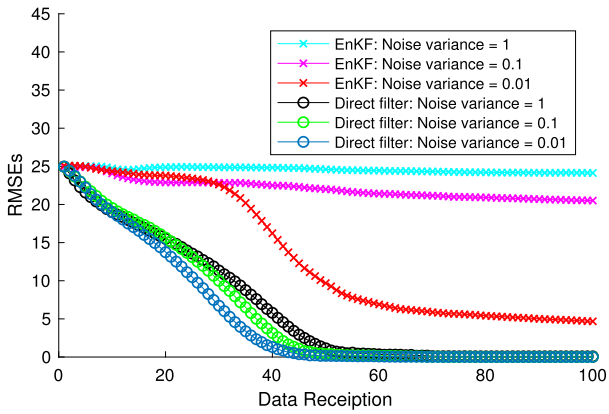


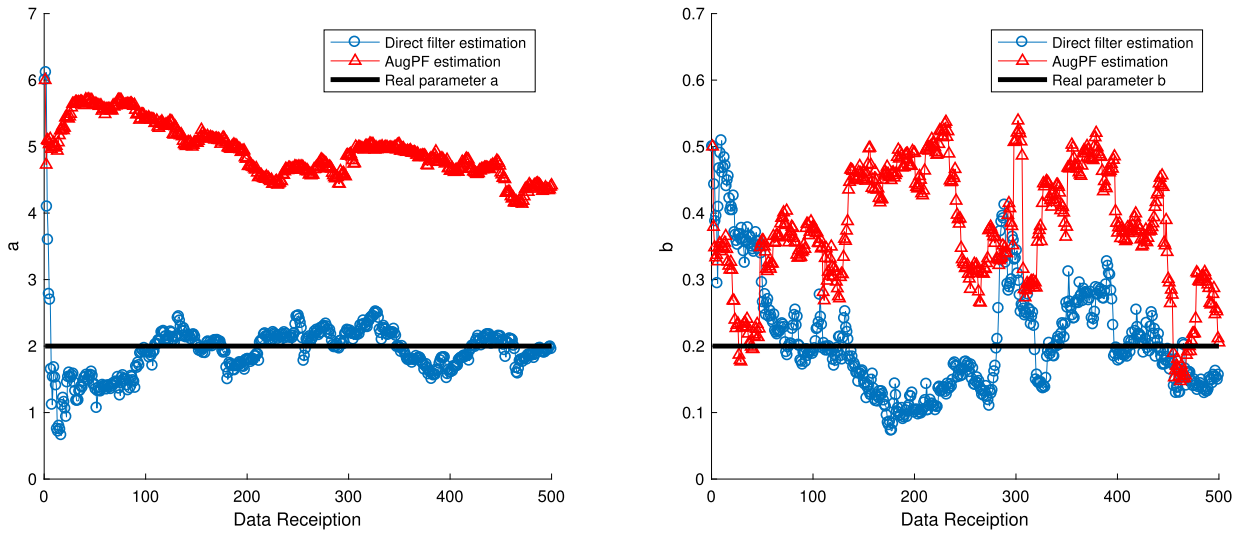
Fig. 9. Comparison of RMSEs between the direct filter and the EnKF with observational noise ξ with variance 0.01, 0.1 and 1.

trend of convergence after receiving approximately 40 sets of observational data, and the direct filter could capture the true parameter accurately after approximately 40 estimation steps, which is much faster and more accurate than both the AugPF and the EnKF.

In the direct filter, we drop the observational error in the identity $X_n = H^{-1}(Y_n - \xi_n)$ and describe the state variable X_n by $H^{-1}Y_n$. In the following experiment, we demonstrate the impact of various observational noises on the direct filter by carrying out the above repeated parameter estimation experiment with different variances of the observational noise. In Fig. 9, we plot the comparison of RMSEs between the direct filter and the EnKF in the cases that the observational noise ξ is a Gaussian random variable with variances 0.01, 0.1 and 1. The cyan, magenta and red curves (marked by crosses) show the RMSEs of the EnKF when the variances of the observational noise are 0.01, 0.1 and 1 respectively; the black, green and blue curves (marked by circles) show the RMSEs of the direct filter when the variances of the observational noise are 0.01, 0.1 and 1 respectively. From this figure, we can see that both the direct filter and the EnKF converge faster when the observational noise is smaller. Although the EnKF reaches its fastest convergence when the variance of the observational noise is 0.01, the direct filter obtains much better parameter estimation results compared with the EnKF even in the case that the observations received by the direct filter are perturbed by larger noises. We also observe that when the variance of the observational noise is 1, which is quite large in this Lorenz 96 model, the direct filter catches the true parameter (the RMSEs decrease to 0) after approximately 50 estimation steps; and when the variance of the observational noise is 0.01, the direct filter obtains the true parameter after approximately 40 estimation steps. This indicates that although larger observational noises make the direct filter harder to find the true parameter, it would not cause substantial influence to the performance of the direct filter in this example.

Example 3

Finally, we study a real application encountered by material scientists. In this problem, electron beams in scanning transmission electron microscope (STEM) is used to observe atom positions on some material surface driven by the energy potential V, which is determined by the material property. The goal of this application is to estimate the energy potential



(a) Parameter a (b) Parameter b

Fig. 10. Example 3: Comparison of parameter estimation between the direct filter and AugPF with 2,000 particles for both methods.

by observing atom positions. With the capability of obtaining the micro-scale level energy potentials, we could carry out more sophisticated quantitative studies of elementary mechanisms of solid-state transformations, and pursue predictive atom-by-atom fabrication [28].

From the physical description of Langevin equation [29], we know that for a given potential V, the corresponding Langevin equation is

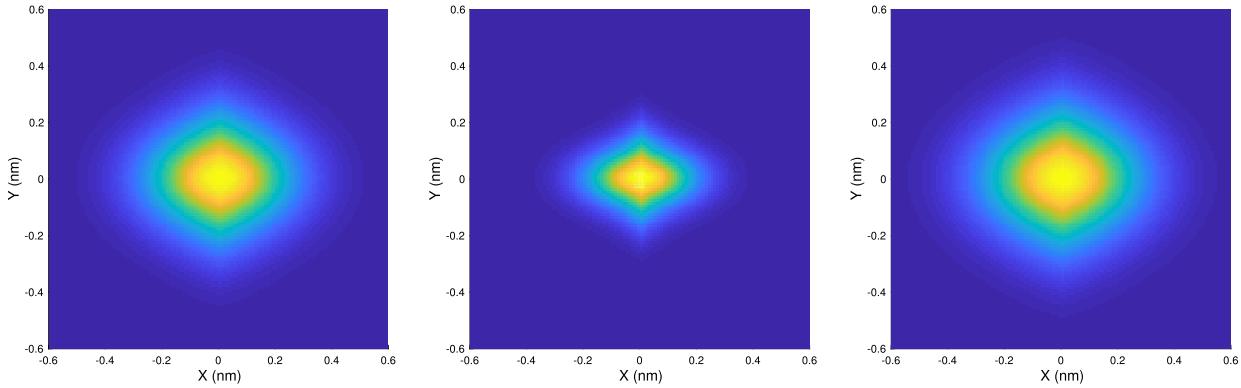
$$dX_t = -\nabla V dt + \sqrt{2D} dW_t, \tag{23}$$

where ∇V is the gradient of the potential function V, W_t is a standard Brownian motion, and D determines the diffusion coefficient, which can be inferred directly based on standard statistical techniques (see [24]). The solution X_t of Langevin equation (23) gives the positions of the target atom at time t and STEM could provide accurate direct observation for X_t. In many situations, we can use physical models to describe the potential V as a function governed by a set of parameters θ and we denote this potential function by V_θ. As an example in this work, we assume that the potential function V_θ is a standard parametric ansatz with the following form [24]

$$V_\theta(x, y) = a(x^2 + y^2)(1 - b\cos(4\arctan(\frac{x}{y}))), \tag{24}$$

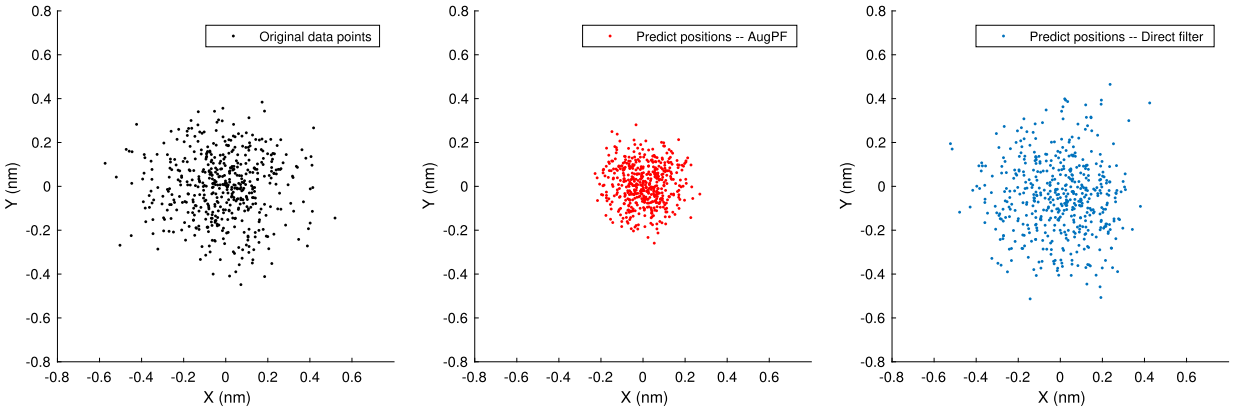
where $X = (x, y)$ is the position of the atom, and the goal is to estimate the parameter pair $\theta = (a, b)$ in the potential model (24) based on observations. The observational data we receive in this problem is the direct observation of atom positions obtained from STEM. The gap between two observations is 0.05 second, which is the practical observation time step of STEM. To formulate a discrete problem, we discretize the Langevin equation (23) with Euler-Maruyama scheme and use the corresponding discrete SDE system as our state model in the parameter estimation problem.

In the numerical simulations, we simulate the Langevin equation with Euler-Maruyama scheme and choose the simulation time step-size as 0.001 second. In this way, we receive one observational data in every 50 model simulations and we want to examine the performance of our direct filter when there's a gap in time step-size between simulation and observation. In this experiment, we process altogether 500 observed atom positions to estimate the parameters a and b in the energy potential model (24), and we calculate the online estimations for the parameters by using the direct filter algorithm and compare with the AugPF due to the relatively higher accuracy of the AugPF in solving low dimensional problems. The observational data are generated synthetically by simulating the state model with pre-chosen "true parameters" $a = 2$ and $b = 0.2$. In Fig. 10, we demonstrate the performance comparison between the direct filter and the AugPF in estimating parameters, and both methods use 2000 particles to implement the particle filter. Subplots 10 (a) and 10 (b) show the estimations for parameters a and b respectively. In each subplot, the straight black line is the true parameter, the red curve marked by triangles shows the estimation obtained by using the AugPF, and the blue curve marked by circles gives the estimation obtained by using the direct filter. We can see from the figure that the direct filter could capture the true parameters in the model more accurately than the AugPF. We then calculate estimations for the invariant parameters from both methods by dropping the first 100 online estimations as burn-in steps and compute the mean values based on the rest 400 online estimations. As a result, we have that the direct filter estimations are $\tilde{a} = 2.0060$, $\tilde{b} = 0.1861$, and the AugPF estimations are $\hat{a} = 4.7989$, $\hat{b} = 0.3897$.



(a) Real energy potential (b) Estimated energy potential – AugPF (c) Estimated energy potential – Direct filter

Fig. 11. Example 3: Comparison of energy potential estimation.



(a) Original data (b) Predicted atom positions – AugPF (c) Predicted atom positions – Direct filter

Fig. 12. Example 3: Comparison between data points and predicted atom positions in the XY-plane.

To demonstrate the performance of parameter estimations in describing the original potential model, which is what we are interested in this application, we plot the approximate potential function V_θ in Fig. 11, by using the estimated parameters obtained from the direct filter and the AugPF, and compare them with the "real potential" generated by true parameters (a, b). Fig. 11 (a) shows the contour graph of the real potential function, Fig. 11 (b) is the contour graph of the approximate potential function obtained from the AugPF, and Fig. 11 (c) is the contour graph of the approximate potential function obtained from the direct filter. We can see that the approximate potential function obtained from the direct filter looks very similar to the real potential, and the approximate potential generated by using the AugPF estimations is much smaller than the real potential. Apparently, the direct filter estimations are more accurate than the AugPF.

A direct consequence of inaccurate energy potential estimation is that it would generate inaccurate future predict atom positions, and being able to predict atom's behavior is crucial in designing and understanding quantum materials. To illustrate the atom position prediction generated by approximate energy potentials, we predict atom positions for the next 500 observation time steps by using potentials derived from different methods and compare the predictions with the original observational data. In Fig. 12 (a), we plot the original observational atom positions; in Fig. 12 (b), we plot the predict atom positions generated by the approximate potential derived from the AugPF; and in Fig. 12 (c), we plot the predict atom positions generated by the approximate potential derived from the direct filter. We can see that the direct filter approximate potential could generate very reliable predict atom positions, which look very similar to the original observational data.

5. Conclusions

In this work, we developed a novel on-line Bayesian parameter estimation method by solving a nonlinear filtering problem. We name this approach the direct filter method due to the fact that we only aim on estimating the parameter as

the target state of the nonlinear filtering problem that we derived from the original parameter estimation problem. The novelty of this direct filter approach arises in the procedure of modeling the parameter that we are interested in as a parameter process and compositing the original observation function with the state model to link the parameter process and the observational data. This direct filter method for parameter estimation is essentially different from other nonlinear filter approaches, such like the augmented filter which combines the parameter process and the dynamical system together as the target state in the corresponding nonlinear filtering problem. This suggests the high efficiency and high accuracy for the direct filter since it reduces the dimension of the target state in the filtering problem and utilizes both the model information and the observational information to estimate only the parameter. Numerical experiments demonstrate the efficiency and accuracy of the direct filter approach in both classic parameter estimation problems and in real application problems.

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